

INTELLIGENT FOOD WASTE PREDICTION AND REDISTRIBUTION SYSTEM

PHASE I REPORT

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BONAFIDE CERTIFICATE

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This project addresses the following Sustainable Development Goal(s):

SDG 2 – Zero Hunger

SDG 12 – Responsible Consumption and Production

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ABSTRACT

Food wastage is a major problem in college cafeterias. The number of students who come for breakfast and lunch changes every day, and the quantity of food is usually decided by experience and rough guesses. Over-preparation leads to cooked food being thrown away, while under-preparation leaves students unserved. At the same time, surplus food that is still safe to eat rarely reaches the people who need it, because there is no organised link between the cafeteria and nearby NGOs.

To address these problems, the Intelligent Food Waste Prediction and Redistribution System is proposed. Students vote on a website to indicate whether they will have breakfast or lunch. The votes are stored in a PostgreSQL database and combined with historical consumption and attendance data, weather information and calendar features such as the day of the week, holidays and examinations. XGBoost and Random Forest models, served through a Python FastAPI service, predict the expected number of students, and the predicted attendance is converted into a recommended quantity of food to prepare. The models are to be evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 , with a chronological train and test split.

After each meal the cafeteria records the quantity prepared and consumed, and the system calculates the surplus and the waste rate. When edible surplus food is available, the nearest registered NGOs are found with a distance-based search and alerted by SMS, push notification or e-mail. The NGO can accept or reject the request, and the pickup and feedback are recorded. An administrator dashboard shows live votes, predictions, surplus records, NGO activity and food-waste reduction reports.

The system is built in three layers: a React and Bootstrap frontend, a PHP backend with a separate machine learning service, and a PostgreSQL data layer. This Phase I report presents the problem analysis, the literature survey, the architecture, the database design and the module design. By bringing demand prediction, surplus monitoring and NGO redistribution into one platform, the system aims to reduce food waste on campus and to support the Sustainable Development Goals of Zero Hunger, Responsible Consumption and Production, and Climate Action. Future versions can add dinner and per-dish prediction, machine vision for plate waste, IoT food monitoring and a mobile application.

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LIST OF ABBREVIATIONS

<i>ABBREVIATION</i>	<i>EXPANSION</i>
AI	Artificial Intelligence
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
DFD	Data Flow Diagram
IoT	Internet of Things
JSON	JavaScript Object Notation
MAE	Mean Absolute Error
ML	Machine Learning
NGO	Non-Governmental Organisation
PDO	PHP Data Objects
PHP	PHP: Hypertext Preprocessor
RF	Random Forest
REST	Representational State Transfer
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SMS	Short Message Service
SQL	Structured Query Language
UI	User Interface
XGBoost	Extreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Every day a college cafeteria has to answer one difficult question: how much food should be cooked? Breakfast and lunch are prepared for hundreds or thousands of students, but the number who actually turn up changes from day to day. It depends on the menu, the weather, the timetable, exams, holidays and many small habits that no one writes down. Cooks therefore decide the quantity from experience and rough guesses. When the guess is too high, cooked food is left over and is usually thrown away. When it is too low, students go hungry and the cafeteria loses their trust.

The cost of this guesswork is large. Food that is thrown away carries the water, energy, labour and money that went into growing, transporting and cooking it, and wasted food that ends in landfill adds to greenhouse gas emissions. At the same time, many people near a campus do not get enough to eat. Surplus food that is still fresh and safe could reach them, but it rarely does, because there is no quick and organised way to tell a charity that food is available.

Recent progress in machine learning, web development and notification services makes it practical to attack both halves of this problem together. A simple website can collect what students plan to eat. A trained model can turn those plans, together with history and weather, into a reliable demand estimate. A database can record what was cooked and what was eaten, and an alert service can pass any real surplus to nearby non- governmental organisations (NGOs). This project joins these pieces into one system, the Intelligent Food Waste Prediction and Redistribution System.

1.2 OBJECTIVE

The main objective of the INTELLIGENT FOOD WASTE PREDICTION AND REDISTRIBUTION SYSTEM is to reduce food wastage in college cafeterias by accurately predicting the daily food requirements using Machine Learning. The system aims to collect student meal attendance preferences through an online voting system and analyze them along with historical food consumption, attendance patterns, weather conditions, weekdays, and other relevant factors. Based on these inputs, the system predicts the expected food demand and helps cafeteria management prepare the appropriate quantity of food. It also aims to identify surplus food and notify registered NGOs or authorized organizations for timely collection and redistribution.

1.3 EXISTING SYSTEM

In the existing system, college cafeterias generally prepare food based on estimated student attendance and previous consumption patterns. The actual number of students attending meals may vary from day to day, which can result in excess food preparation and wastage. Most cafeterias do not have an intelligent system to collect student meal preferences or predict food demand using multiple factors such as historical consumption, weather conditions, weekdays, and attendance patterns. Surplus food is often managed manually, and there may be no automated mechanism to identify excess food and notify NGOs for timely redistribution. This traditional approach can lead to inefficient resource utilization, increased food wastage, and difficulty in managing surplus food effectively. Therefore, there is a need for an intelligent and automated system that can predict food requirements and support efficient redistribution of excess food.

1.4 PROPOSED SYSTEM

The proposed INTELLIGENT FOOD WASTE PREDICTION AND REDISTRIBUTION SYSTEM is an AI-based system designed to predict food demand and reduce food wastage in college cafeterias. The system allows students to provide their meal preferences through an online voting system for breakfast and lunch. The collected voting data is combined with historical food consumption, attendance patterns, weather conditions, weekdays, and other relevant factors to create a dataset for Machine Learning. The trained model predicts the expected number of students and the required quantity of food for each meal, helping cafeteria management prepare food more efficiently. When surplus food is detected, the system generates an alert to registered NGOs or authorized organizations for timely collection and redistribution. The system also provides an admin dashboard to monitor student votes, food demand, predictions, and surplus food. Thus, the proposed system helps minimize food wastage, improve resource utilization, and promote sustainable food management.

In addition, the system provides an automated and data-driven approach instead of relying only on manual estimation. By integrating student voting, data processing, Machine Learning prediction, surplus detection, and NGO notification, the system creates an effective connection between the cafeteria and food redistribution organizations. This helps ensure that usable surplus food is managed efficiently and can reach people in need.

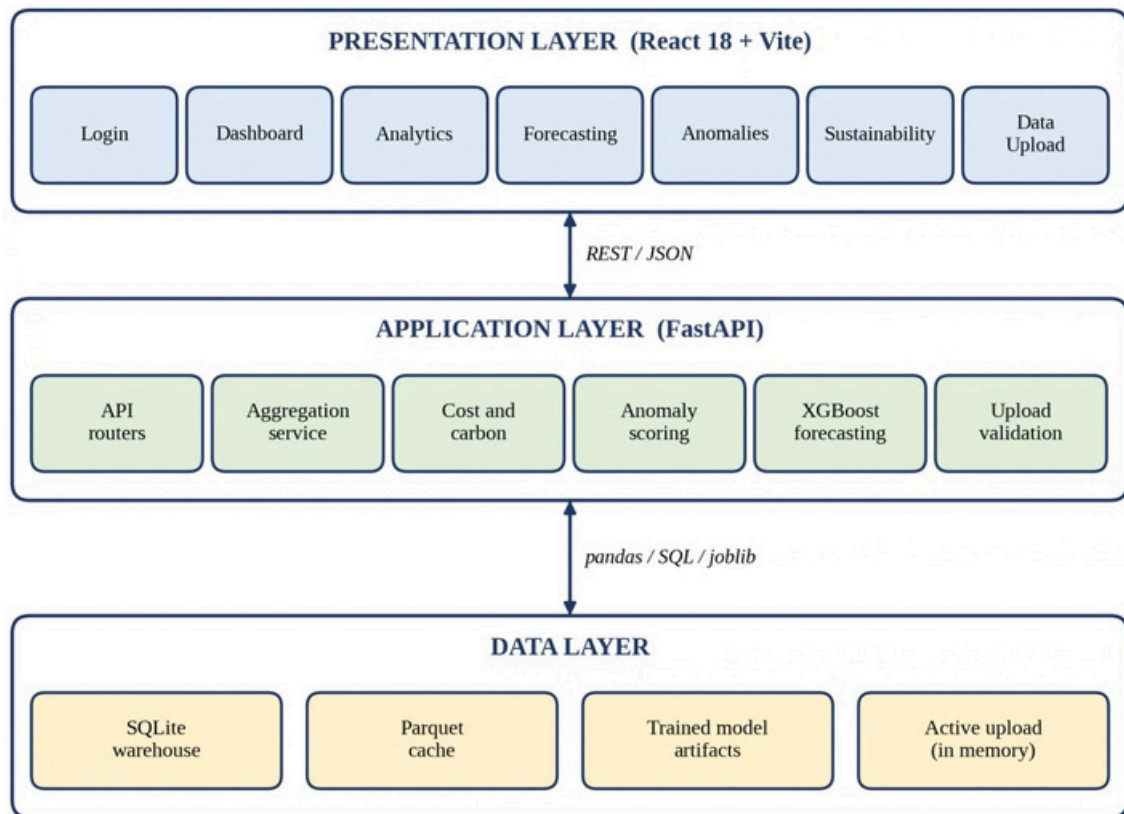


Figure 1.1. Overview of the Proposed System Architecture

CHAPTER 2

LITERATURE SURVEY

This chapter reviews earlier work related to the aims of this project: food-waste prediction, demand and attendance forecasting for cafeterias and catering services, and digital platforms for surplus-food redistribution. The review gives the background for the design choices made in the system.

2.1 LITERATURE REVIEW

2.1.1 Food-Waste Prediction and Machine Learning

Kim and Yoon [3] use machine learning to predict the quantity of food waste in military dining facilities, which are large-scale food-service operations. Their work shows that waste in a big kitchen follows patterns that a model can learn, and that these predictions can be used for planning. Wan et al. [2] apply a machine learning framework to predict compost maturity in an automated food-waste composting system, which shows that machine learning is also useful in handling waste after it has been produced. Kumar et al. [5] combine a Random Forest model with a web application, geolocation and automated notifications to predict food waste and to connect it with nearby NGOs. This is the closest to the aim of the present project, and it supports the choice of tree-based models and of a distance-based search for NGOs.

2.1.2 Demand and Attendance Forecasting for Cafeterias

Eriksson et al. [6] study machine learning methods to plan guest attendance in Swedish public catering, and report that better attendance planning lowers food waste and saves money. Rodrigues et al. [7] build models for short-term demand forecasting in food catering services using historical catering data. Turker [8] uses campus-related data and regression models to predict daily meal demand in campus dining, and shows how such predictions support sustainability. Faezirad et al. [9] use an artificial neural network on meal reservation data to model student show and no-show behaviour in subsidy-based university dining. This is important for the present project because a vote or a reservation is not the same as an actual visit, and the model must learn the gap between them. Jung and Kim [10] develop predictive models, including XGBoost and Random Forest, together with a web application for university cafeteria attendance, and show that breakfast, lunch and dinner attendance can be predicted. Oroski and da Silva [15]

use machine learning demand forecasting in a bakery case study and also assess the environmental benefits through a life-cycle analysis.

The common conclusion of these studies is that data-driven forecasting of demand reduces waste. They mostly rely on historical attendance or bookings, and few of them let students state their requirement directly through a voting interface that feeds the model.

2.1.3 Surplus-Food Redistribution Platforms

Suresh et al. [4] present ZeroHunger, a digital solution that combines AI-based food donation management with real-time coordination between donors and organisations. Devika et al. [11] describe a web-based food donation system with a central database, location tracking and automated notifications to support NGOs. Agmapisarn et al. [12] propose a smart food-waste rescue hub that uses IoT sensors, Firebase and NGO notification to monitor food conditions and coordinate redistribution. Murthy et al. [18] propose a web system that records surplus food information and sends alerts to nearby NGOs, volunteers and other collection organisations. Shivani and Usha [14] stress that real-time information, accessibility and charitable collaboration are important for effective surplus-food sharing. Hübner et al. [17] review food-waste-reducing platforms and find that digital platforms can connect surplus suppliers with recipients, monitor waste and generate useful data. Tanksale et al. [16] design an integrated foodbank network using mixed-integer programming, with a case study in Delhi, India. Shi et al. [1] study bandit-based optimisation for crowdsourced food-rescue platforms and show how data-driven methods improve food-rescue operations and volunteer engagement.

Li et al. [13] take a different route and use machine vision and deep learning to identify food waste on plates in university cafeterias. This is a possible future extension of the present work, since the current system measures surplus from recorded quantities and not from images.

2.1.4 Techniques Used in This Project

Chen and Guestrin [19] introduced XGBoost, a scalable tree-boosting system that trains quickly and handles structured data well. Breiman [20] introduced Random Forests, which average many decision trees to reduce variance and give stable predictions on tabular data. Both are used here and compared. Pedregosa et al. [21] describe scikit-learn, earn,

learn, which supplies the preprocessing tools and the evaluation metrics. Hyndman and Athanasopoulos [23] point out that a forecasting model must be tested on data that comes after the training period; this is the reason for the chronological train and test split planned in this project. The distance between the cafeteria and an NGO is calculated with the haversine formula [22], which gives the great-circle distance between two points from their latitude and longitude.

2.2 COMPARISON AND DISCUSSION

Table 2.1 groups the reviewed studies by their focus and shows which part of the problem each of them covers.

Table 2.1. Summary of the reviewed literature by focus area

<i>Focus area</i>	<i>Representative studies</i>	<i>What they contribute</i>
Food-waste prediction	Kim and Yoon [3]; Wan et al. [2]; Kumar et al. [5]	ML can predict waste quantity and waste-processing outcomes; tree-based models work well
Cafeteria demand and attendance forecasting	Eriksson et al. [6]; Rodrigues et al. [7]; Turker [8]; Faezrad et al. [9]; Jung and Kim [10]	Attendance and demand can be predicted from history, reservations and campus data, and doing so reduces waste
Redistribution platforms	Suresh et al. [4]; Devika et al. [11]; Agmapisarn et al. [12]; Murthy et al. [18]; Hübner et al. [17]	Web and mobile platforms with location tracking and alerts connect donors with NGOs
Network and operations	Tanksale et al. [16]; Shi et al. [1]	Optimisation of foodbank networks and volunteer engagement
Sensing and vision	Li et al. [13]; Agmapisarn et al. [12]	Cameras and IoT sensors can measure waste and food condition automatically

Looking back at Section 2.1, each group of studies deals with one piece of the problem. Forecasting papers explain how to predict how many people will eat, platform papers explain how to hand surplus food to charities, and the techniques papers explain how to build and evaluate the models. What is less common is one workflow that begins with the students' own votes, predicts demand with machine learning, records the real surplus after the meal and then alerts the nearest NGOs. The proposed system combines these steps in a single application and keeps a feedback loop, because the recorded actual consumption becomes new training data.

CHAPTER 3

SYSTEM DESIGN

This chapter explains how the system is designed, using diagrams to show the workflow and the way the components talk to each other. It also covers the software, hardware and dataset requirements, the deployment setup, the activity diagram, the data flow diagram and the database design.

3.1 SYSTEM ARCHITECTURE

The system is built as a web application for three kinds of users: students, cafeteria staff or administrators, and NGOs. Figure 3.1 shows the complete architecture. It has five parts: the frontend, the PHP backend, the PostgreSQL database, the machine learning service and the external services.

Frontend. The frontend uses React with Bootstrap, so that the pages are responsive and work on mobile phones. The student interface allows registration and login, voting for breakfast or lunch, viewing today's menu, viewing surplus alerts and viewing the vote history. The NGO interface allows registration and login, viewing surplus alerts, accepting or rejecting requests and viewing the pickup history. The administrator interface shows live votes and predictions, meal planning and surplus, NGO management, reports and analytics, and system settings.

PHP backend. The backend has an API layer that offers RESTful endpoints, handles authentication and authorisation, and validates requests. The business logic layer processes votes, handles prediction requests, runs the surplus detection logic, matches NGOs and triggers notifications. The data access layer uses PHP Data Objects (PDO) for CRUD operations, a query builder and database transactions. A file and model handler loads the trained model and sends data to the Python service.

Database. PostgreSQL stores users, meals, votes, surpluses and NGOs. The tables are described in Section 3.3.3.

Machine learning service. A Python service built with FastAPI exposes a prediction endpoint (POST /api/predict). It receives the features, loads the trained model and returns the predicted demand. The models are XGBoost and Random Forest. Feature engineering uses weather, day of the week, exam or holiday status, historical votes and past consumption. During training the models are evaluated with Mean Absolute Error (MAE) and the best one is saved as a pickle file.

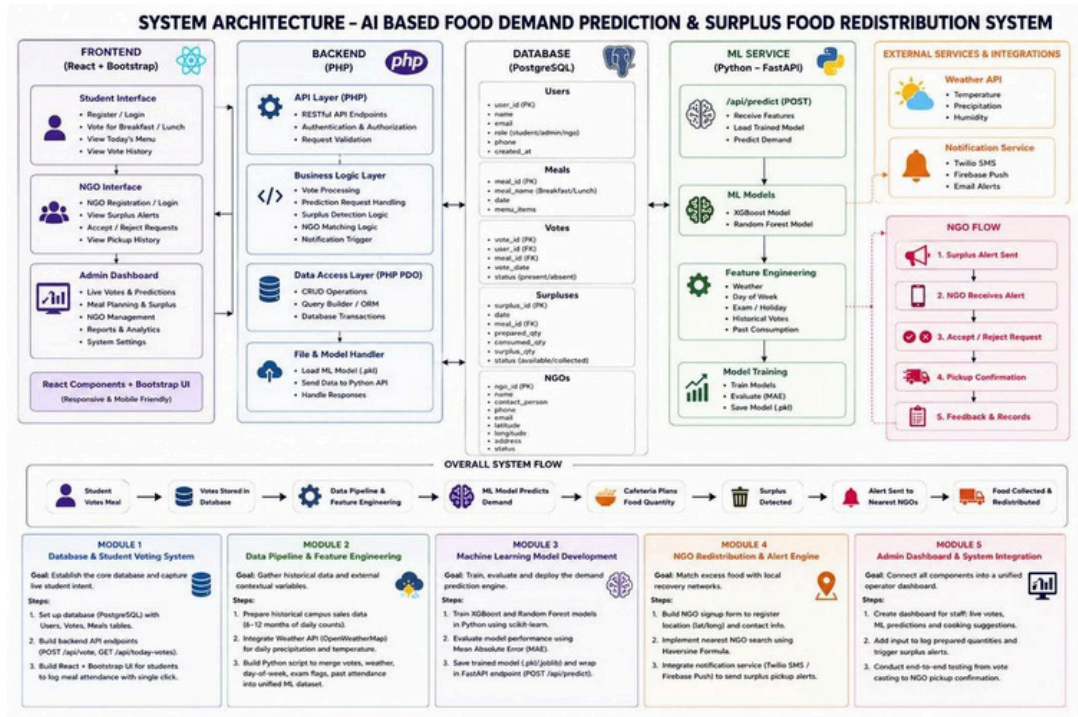


Figure 3.1. Architecture diagram for the entire system

3.2 SYSTEM REQUIREMENTS

This section lists what the system needs in terms of software, hardware, data and deployment.

3.2.1 SOFTWARE REQUIREMENTS

The frontend is written in React and styled with Bootstrap. The backend business logic is written in PHP and uses PDO to talk to PostgreSQL. The machine learning part is written in Python, where Pandas and NumPy handle the data, scikit-learn supplies the preprocessing and metrics, XGBoost and Random Forest train the models and joblib stores them. FastAPI serves the trained models. Table 3.1 lists the components and their purpose.

Table 3.1. Software requirements of the proposed system

<i>Software component</i>	<i>Suggested version</i>	<i>Purpose</i>
React.js / Bootstrap	React 18.x, Bootstrap 5.x	Responsive user interface for students, administrators and NGOs
PHP	8.1 or higher	Backend REST API, authentication
PostgreSQL	14 or higher	Relational database for users, meals, votes, surpluses and NGOs
Python	3.10 or higher	Machine learning service and data pipeline
FastAPI / Uvicorn	Latest stable release	Serving the trained model through a prediction endpoint
Pandas / NumPy	Latest stable release	Data cleaning, feature engineering and analysis
scikit-learn / XGBoost	Latest stable release	Random Forest, XGBoost models, encoders and evaluation metrics
joblib	Latest stable release	Saving and loading trained model files
Weather API	OpenWeatherMap or equivalent	Temperature, precipitation and humidity for each day
Twilio / Firebase / SMTP	Service accounts	SMS, push and e-mail alerts to NGOs
Git / Postman / pytest	Latest stable release	Version control, API testing and automated testing

3.2.2 HARDWARE REQUIREMENTS

The models are small tabular models, so training and prediction both run on an ordinary CPU and no GPU is needed. Table 3.2 gives suggested values for development and demonstration. Students and NGOs only need a device with a web browser, such as a phone or a laptop

Table 3.2. Hardware requirements of the proposed system

<i>Hardware component</i>	<i>Suggested specification</i>	<i>Purpose</i>
CPU	Multi-core processor with 4 or more cores	Serving requests and training models
RAM	8 GB or higher	Database, ML service and web server running together
Storage	SSD with sufficient free space	Database, historical data and model files
GPU	Not required	Models run on the CPU

3.2.3 DATASET REQUIREMENTS

The prediction models need history. The following data items are required to develop, train and test the system.

- Daily breakfast and lunch attendance of the cafeteria for at least six to twelve months, with the quantity prepared and the quantity consumed.
- Student voting records: how many students voted for each meal, and the time of the vote.
- Calendar information: day of the week, holidays, examination days and special events.
- Weather data for each day: temperature, precipitation and humidity.
- Menu information, so that the effect of popular and unpopular dishes can be

3.2.4 DEPLOYMENT AND SCALING

In the first version, the website, the PHP backend, the Python service and the database run on a single server inside the campus or on a small cloud instance. The Python service is called only over HTTP by the PHP backend, so it can be moved to a separate machine later. Model files are loaded when the service starts and are replaced only after retraining. For many campuses or for much heavier traffic, the database can be moved to a managed service, the ML service can be run in several instances behind a load balancer, and the notification calls can be placed in a queue.

3.3 SYSTEM DESIGN

The diagrams in this section show how the system works and how its components communicate

3.3.1 ACTIVITY DIAGRAM

The activity diagram in Figure 3.2 traces the steps of the system. The flow starts when a student logs in and votes for breakfast or lunch before the deadline. The votes are stored and the system collects the features for that meal. The model predicts the attendance and the cafeteria prepares food accordingly. After the meal the staff record the prepared and consumed quantities. If edible surplus remains, the system finds the nearest registered NGOs and sends them an alert. An NGO can accept and collect the food and give feedback; if there is no surplus, a zero record is stored. The administrator dashboard is updated in both cases, and the recorded actual values feed the next round of model training.

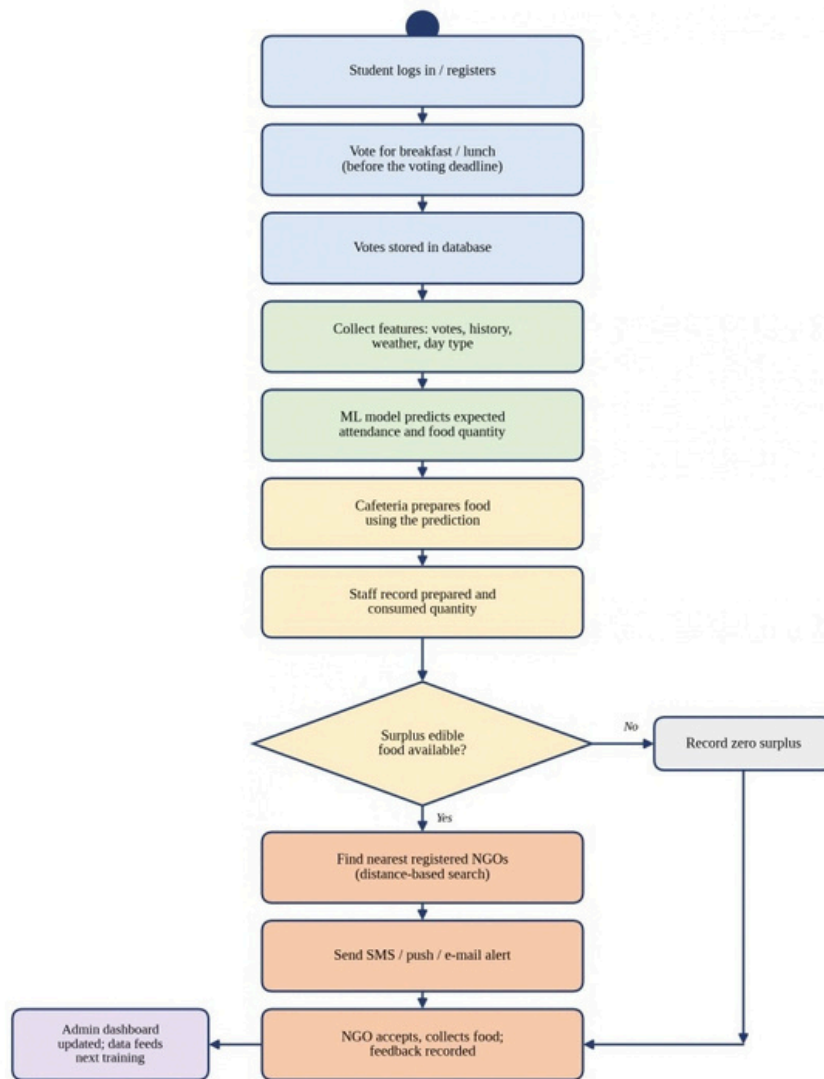


Figure 3.2. Activity diagram of the System

3.3.2 DATA FLOW DIAGRAM

The data flow diagram shows how data moves between the components. Figure 3.3 is the level 0 diagram. Students send their meal votes to the system. The cafeteria staff or administrator send meal details and the prepared and consumed quantities and receive predictions and reports. NGOs register, receive surplus alerts and send back their acceptance or rejection. Internally the system reads and writes the user, meal and vote data (D1), the surplus and NGO records (D2) and loads the trained models (D3).

3.3.2 DATA FLOW DIAGRAM

The data flow diagram shows how data moves between components. Figure 3.3 is the level 0 diagram: the user sends filters, requests and CSV files to the platform and gets back dashboards, forecasts, anomalies and reports.

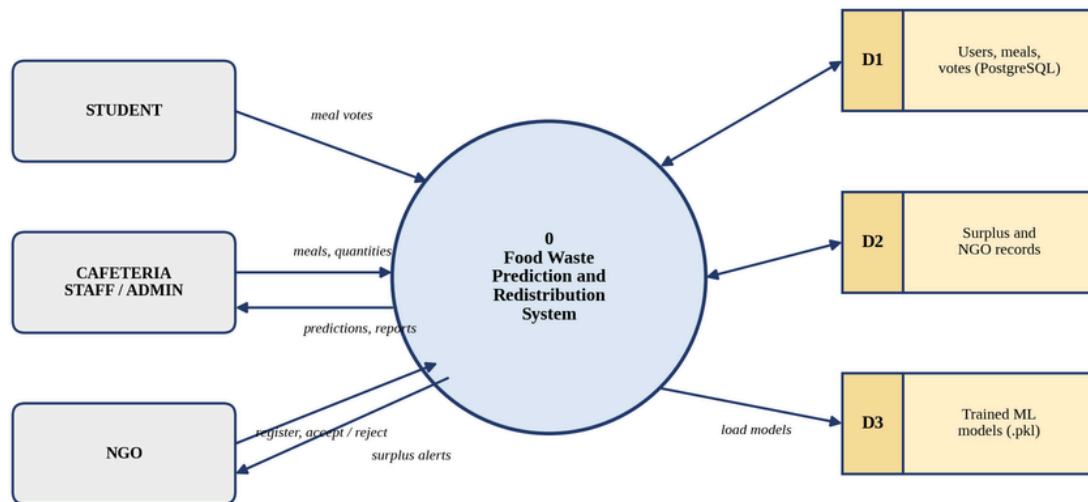


Figure 3.3. Data Flow diagram of the System

3.3.3 DATABASE DESIGN

The PostgreSQL database has five main tables, shown in Figure 3.4 and summarised in Table 3.3. A user can be a student, an administrator or an NGO representative. A meal is a breakfast or lunch on a particular date. A vote links a student to a meal. A surplus record belongs to a meal and stores the prepared, consumed and surplus quantities. An NGO record stores the contact person and the latitude and longitude that are needed for the distance search.

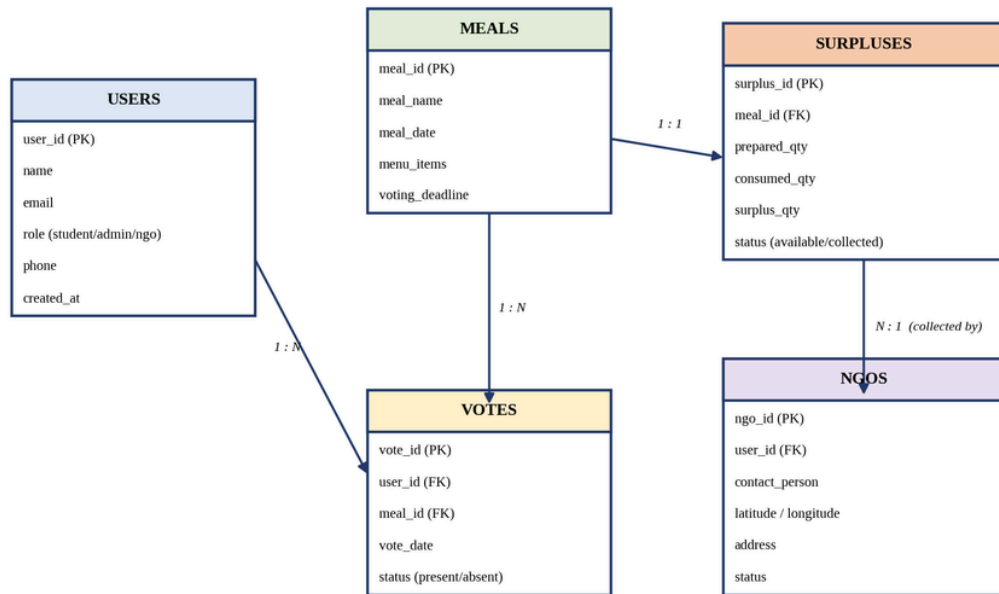


Figure 3.4. Database design of the System

Table 3.3. Summary of the database tables

<i>Table</i>	<i>Main attributes</i>	<i>Purpose</i>
users	user_id, name, email, role, phone, created_at	Stores students, administrators and NGO representatives with their role
meals	meal_id, meal_name, meal_date, menu_items, voting_deadline	Stores each breakfast and lunch that students can vote for
votes	vote_id, user_id, meal_id, vote_date, status	Stores each student's vote and later whether the student was present
surpluses	surplus_id, meal_id, prepared_qty, consumed_qty, surplus_qty, status	Stores the quantities for each meal and whether surplus is available or collected
ngos	ngo_id, user_id, contact_person, latitude, longitude, address, status	Stores registered NGOs and their location for the nearest-NGO search

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

This section describes the modules of the system, what happens inside them and how they interact. The system works on student votes, historical cafeteria records, calendar and weather data, and the surplus records created after each meal. The approach in brief is: student voting data, historical food consumption and attendance data are given to an AI/ML prediction model, which produces a forecast of future meal demand. The forecast guides the quantity cooked, and the difference between cooked and consumed food is handled as surplus.

To keep the work manageable, the system is divided into five modules, and each module has a clear goal. The modules talk to each other through database tables and REST endpoints, so each of them can be developed and tested on its own.

4.2 MODULES

The system has five modules.

- Database and Student Voting System
- Data Pipeline and Feature Engineering
- Machine Learning Model Development
- NGO Redistribution and Alert Engine
- Admin Dashboard and System Integration

4.2.1 DATABASE AND STUDENT VOTING SYSTEM

The goal of this module is to set up the core database and to capture the live intention of students.

Database Setup Module: This module creates the PostgreSQL database with the users, meals, votes, surpluses and NGOs tables, the primary and foreign keys and the indexes that are needed for fast daily queries.

Voting API Module: The PHP backend offers endpoints to post a vote, to read today's votes and to read a student's vote history. Each student can vote once for each meal. The module checks that the voting deadline has not passed and rejects duplicate votes.

Voting Interface Module: A React and Bootstrap page lets the student log in, see today's menu and mark attendance for breakfast or lunch with a single tap. The page is designed for mobile phones, since most students will use it from a phone.

4.2.2 DATA PIPELINE AND FEATURE ENGINEERING

This module prepares the data that the models learn from.

Historical Data Module: This module gathers historical campus consumption data for six to twelve months of daily records, cleans it, removes duplicates and fixes missing values.

Weather Module: This module calls a weather API for the daily temperature, precipitation and humidity and stores the values with the date.

Feature Builder Module: A Python script merges the votes, history, weather and calendar data into one unified dataset. The main features are the number of votes for the meal, the day of the week, whether the day is a holiday or an examination day, the temperature, the precipitation, the attendance on the previous day and on the same day of the previous week, and the average of the last seven days. A very useful feature is the show-up ratio of past meals, given by equation 4.1.

$$(4.1) \quad \alpha = A / V$$

In equation 4.1, A is the number of students who actually ate the meal and V is the number who voted for it. If students are inclined to vote and then not come, α is below one; the model learns this behaviour from the history instead of trusting the raw vote count.

4.2.3 MACHINE LEARNING MODEL DEVELOPMENT

This module trains, evaluates and deploys the demand prediction models.

Training Module: XGBoost and Random Forest regressors are trained on the unified dataset to predict the number of students for a meal, as in equation 4.2. The data is split in time order, with the earlier part used for training and the later part for testing, so that the model is always tested on days that come after the days it learned from.

$$(4.2) \quad \hat{A} = f(V, H, W, C)$$

Here $\hat{A}$ is the predicted attendance, V is the vote count, H is the historical consumption, W is the weather and C is the calendar information. The model with the better test result is saved as a pickle file with joblib.

Prediction Service Module: A FastAPI service loads the saved model and exposes POST /api/predict. It receives the features for a meal and returns the predicted attendance. The PHP backend calls this endpoint when the administrator opens the planning page.

Quantity Recommendation Module: The predicted attendance is converted into a quantity to prepare using equation 4.3, where p is the average portion size per student and β is a small safety buffer set by the cafeteria, for example five percent. The buffer protects against the prediction error, so that a hungry student is not turned away.

$$(4.3) \quad Q = \hat{A} \times p \times (1 + \beta)$$

Evaluation Metrics: The models will be evaluated on the held-out later part of the data. Mean Absolute Error gives the average size of the error and is shown in equation 4.4. Root Mean Squared Error gives more weight to large errors and is shown in equation 4.5. The coefficient of determination shows how much of the variation the model explains and is shown in equation 4.6.

$$MAE = (1/n) \sum |y_i - \hat{y}_i| \quad (4.4)$$

$$RMSE = \text{sqrt}((1/n) \sum (y_i - \hat{y}_i)^2) \quad (4.5)$$

$$R^2 = 1 - \sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2 \quad (4.6)$$

4.2.4 NGO REDISTRIBUTION AND ALERT ENGINE

The goal of this module is to match excess food with local recovery networks. **Surplus Detection Module:** After each meal the cafeteria staff enter the prepared and the consumed quantity. The system calculates the surplus S and the waste rate with equations 4.7 and 4.8, where Q_{prep} is the quantity prepared and Q_{cons} is the quantity consumed. If the surplus is above a configured minimum and the food is marked as edible and within its safe holding time, the record is given the status available.

$$(4.7) \quad S = Q_{prep} - Q_{cons}$$

$$(4.8) \quad \text{Waste rate (\%)} = (S / Q_{prep}) \times 100$$

NGO Registration Module: NGOs sign up with contact details, address and location, which are saved with their latitude and longitude.

Nearest NGO Search Module: The distance between the cafeteria and each registered NGO is calculated with the haversine formula in equation 4.9, where ϕ is latitude, λ is

longitude and R is the radius of the earth. The NGOs are sorted by distance and the nearest ones are chosen.

$$d = 2R \cdot \arcsin(\sqrt{\sin^2(\Delta\phi/2) + \cos \phi_1 \cos \phi_2 \sin^2(\Delta\lambda/2)}) \quad (4.9)$$

Alert Module: The selected NGOs are notified through SMS, push notification and e-mail. The alert states the meal, the approximate quantity, the pickup time window and the location. The NGO can accept or reject the request from its portal. If the first NGO rejects or does not answer within a set time, the next nearest NGO is alerted.

Pickup Tracking Module: When an NGO accepts, the surplus record changes to collected after the pickup is confirmed, and the NGO can leave feedback. This closes the loop and gives the college a record of the food that reached people in need.

4.2.5 ADMIN DASHBOARD AND SYSTEM INTEGRATION

This module joins all components into one unified operator dashboard.

Dashboard Module: The dashboard shows live votes, the predicted attendance and the recommended quantity, the surplus records and the NGO status. Filters allow the administrator to look at a date range, a meal type or an NGO.

Reports and Analytics Module: This module shows food consumption trends, surplus analysis and waste-reduction metrics, and allows the reports to be exported. The reduction in the waste rate is calculated by equation 4.10, where WR_0 is the waste rate before the system was used and WR_1 is the waste rate after it.

$$(4.10) \quad \text{Reduction (\%)} = ((WR_0 - WR_1) / WR_0) \times 100$$

Integration and Testing Module: This module links the frontend, the backend, the database and the ML service, adds input validation and error handling, and runs end-to-end tests from a vote to an NGO pickup confirmation.

4.3 COMPLETE WORKFLOW ANALYSIS

This section follows the complete workflow of the system from a vote to a pickup. There are two paths, as Figure 4.1 shows. The planning path runs before each meal and decides how much food to cook. The redistribution path runs after each meal and handles any surplus. The recorded actual values from the second path become training data for the first.

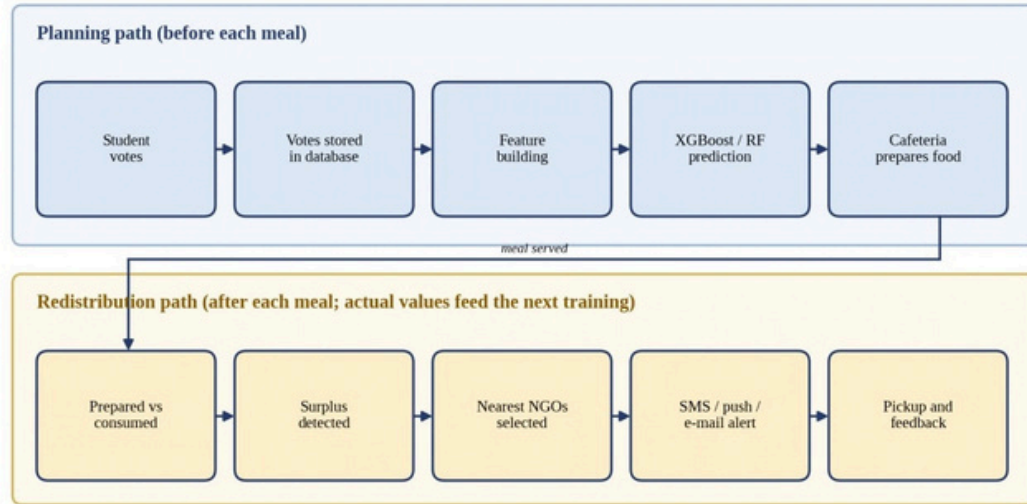


Figure 4.1. End-to-end workflow of the platform

4.3.1 STAGE-BY-STAGE ANALYSIS

The workflow has eight stages. The first two take place before the meal, the next three at planning time, and the rest after the meal. Table 4.1 lists the stages and what happens in each.

Table 4.1. Stages of the complete workflow

#	Stage	Process	Key component
1	Voting	Student logs in and votes for breakfast or lunch before the deadline	React voting page, PHP API
2	Vote storage	The vote is validated and stored once per student per meal	PostgreSQL votes table
3	Feature building	Votes, history, weather and calendar data are merged into model features	Python data
4	Prediction	Model returns predicted attendance; quantity to prepare is calculated	pipeline FastAPI, XGBoost
5	Cooking and recording	Cafeteria prepares the food and later records the consumed quantity	/ RF
6	Surplus detection	Surplus and waste rate are calculated; edible surplus is marked available	Admin dashboard Surplus logic in backend
7	NGO alert	Nearest NGOs are found and alerted; the next NGO is tried on rejection	Haversine search, SMS / push / e-mail
8	Pickup and feedback	NGO accepts, collects and gives feedback; records feed analytics and retraining	NGO portal, reports

4.3.2 COMPARISON WITH THE MANUAL SYSTEM

The proposed system differs from the manual practice in how demand is estimated, how surplus is handled and what is recorded. Table 4.2 compares them. The comparison brings out the main design choice, which is to replace guesswork with a prediction that is checked against the actual result every day.

Table 4.2. Comparison of the manual system and the proposed system

<i>Attribute</i>	<i>Manual system</i>	<i>Proposed system</i>
Demand estimate	Experience and rough attendance	Votes plus history, weather and calendar through an ML model
Quantity decision	Cook decides by habit	Predicted attendance $\times$ portion $\times$ safety buffer
Surplus record	Not recorded or recorded on paper	Recorded in the database after every meal
NGO contact	Informal, often too late	Automatic alerts to the nearest registered NGOs
Learning over time	None	Actual values are added to the training data
Reporting	Manual, occasional	Dashboard with trends and waste-reduction metrics

4.3.3 STATE AND CONSISTENCY ANALYSIS

Several points need careful handling so that the data stays consistent and the predictions stay useful.

- **Votes are not attendance.** A student may vote and not come, or come without voting. The model is trained against the actual consumption and not the raw vote count, and the show-up ratio in equation 4.1 is used as a feature. A cafeteria can also run for some time with the vote count alone as a fallback.
- **Voting deadline.** Votes must close early enough for the kitchen to act. A vote is accepted only before the deadline, and each student has only one vote per meal.
- **Cold start.** At the start there are few votes and little history. The system will begin with the historical attendance and gradually give more weight to the votes as data accumulates.

- **Food safety.** Only food that has not been served, has been kept at a safe temperature and is within its holding time may be offered for redistribution. The staff mark the surplus as edible, and the alert carries a pickup time limit. ***Single source of truth.***
 - The PostgreSQL database is the only permanent store. The ML service does not keep state apart from the model files, so a restart of the service does not lose any record.
- Privacy.** Only the data that is needed is stored about students. Votes are used in
- aggregate for prediction, and NGOs see only the surplus details and not student information.

4.3.4 PERFORMANCE AND RELIABILITY OBSERVATIONS

The models are small and the prediction for one meal is a single call, so the response time is expected to be low. The voting load is concentrated in a short window before the deadline, which the database handles with indexes and short transactions. If the ML service is not available, the backend falls back to the vote count multiplied by the historical show-up ratio, so the planning page keeps working. If a notification channel fails, the alert is tried on the other channels and the failure is logged. Models will be retrained at regular intervals, and their errors will be monitored so that a drop in accuracy is noticed early.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION PLAN AND EVALUATION SETUP

This is the Phase I report, which presents the problem analysis, the literature survey and the design of the system. The implementation is planned as described in this chapter. No measured results are claimed here; the tables for model performance are prepared and will be filled in after the models are trained and tested in the next phase.

5.1.1 IMPLEMENTATION OF THE VOTING SYSTEM AND DATABASE

The first step is to create the PostgreSQL database and the PHP REST endpoints, and then the React pages for login and voting. Table 5.1 describes the dataset that will be assembled for model development.

Table 5.1. Planned dataset profile

<i>Attribute</i>	<i>Planned value</i>
Source	Cafeteria attendance and consumption records, student votes; synthetic data if real data is insufficient (to be labelled)
Time span	Six to twelve months of daily records
Meals	Breakfast and lunch
Target variable	Number of students who ate the meal (actual attendance)
Main features	Vote count, show-up ratio, day of week, holiday or exam flag, temperature, precipitation, humidity, previous-day and previous-week attendance, seven-day average
Train / test split	Chronological, with the later part kept for testing (for example 80 / 20)

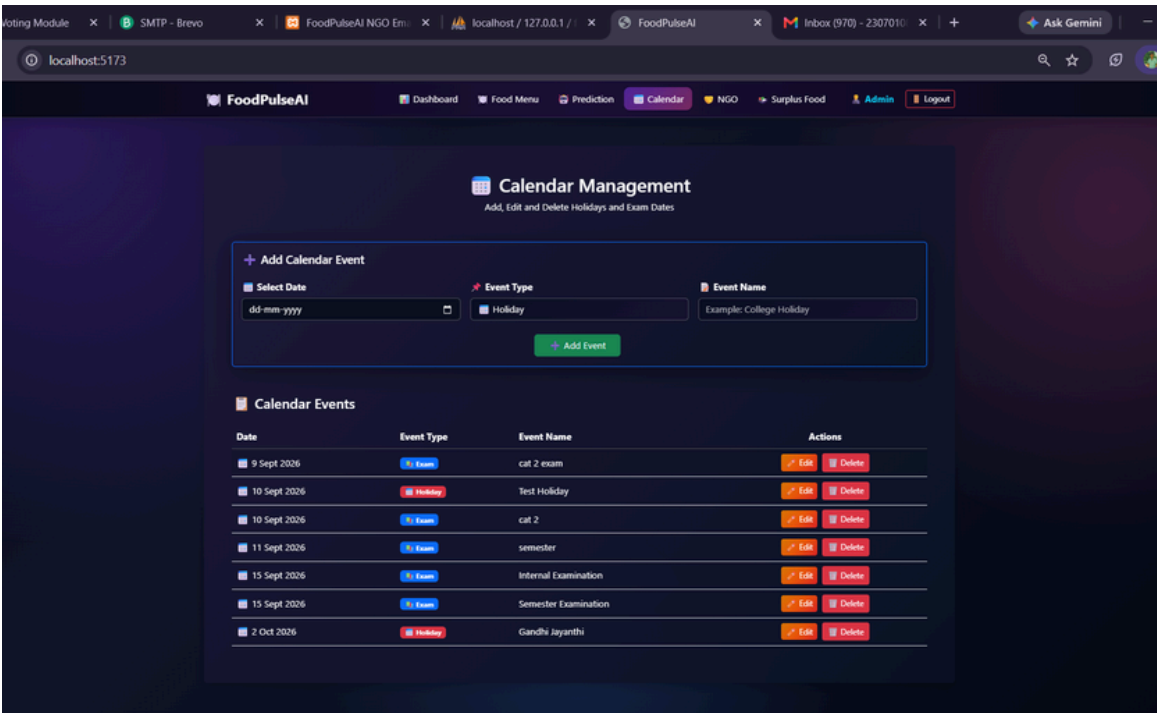
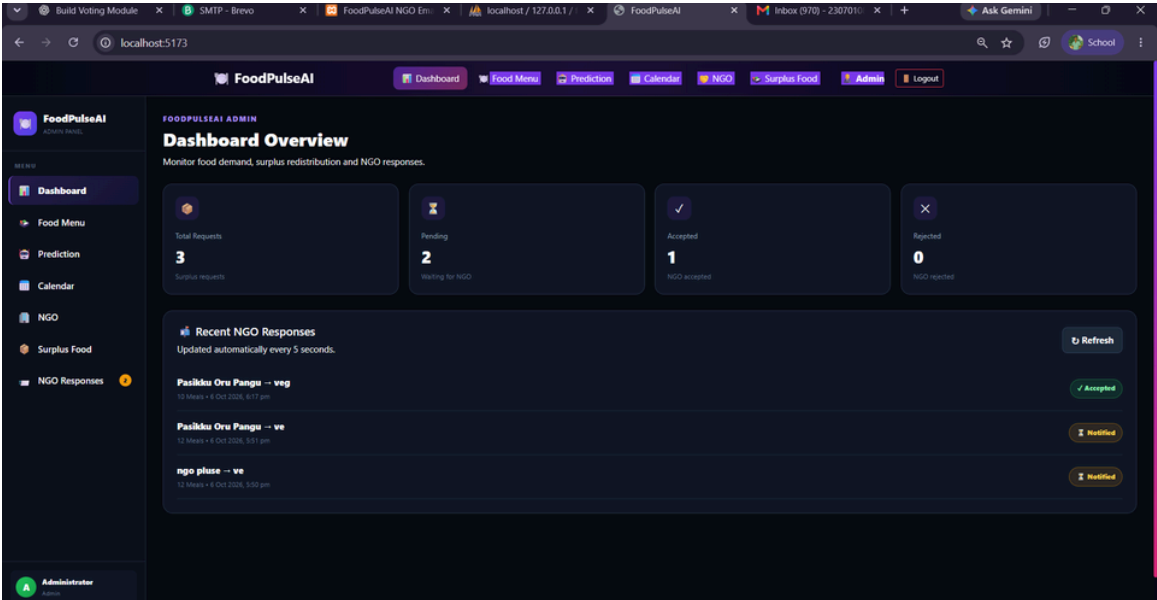
5.1.2 IMPLEMENTATION OF THE FORECASTING MODELS

XGBoost and Random Forest models will be trained on the unified dataset and compared with a simple baseline that multiplies the vote count by the average show-up ratio. The evaluation uses MAE, RMSE and R^2 . Table 5.2 is the template for the results and is left empty on purpose until training is complete.

Table 5.2. Model evaluation template (to be completed after training)

<i>Model</i>	<i>MAE</i>	<i>RMSE</i>	<i>R²</i>
Baseline (votes $\times$ show-up ratio)	—	—	—
Random Forest	—	—	—

The memory use grows with the cache size plus the active upload, and each worker process holds its own copy.



5.1.3 IMPLEMENTATION OF THEREST API

The backend will expose the endpoints listed in Table 5.3. Each endpoint checks the role of the caller, so that only students can vote, only administrators can enter quantities and only NGOs can accept requests.

Table 5.3. Planned REST API endpoints

<i>Method</i>	<i>Endpoint</i>	<i>Purpose</i>
POST	/api/auth/login	Log in a student, administrator or NGO
POST	/api/votes	Submit a vote for a meal before the deadline
GET	/api/votes/today	Live vote counts for today's meals
POST	/api/predict	Predict attendance and quantity (ML service)
POST	/api/surplus	Record prepared and consumed quantity
GET	/api/ngos/nearest	List the nearest NGOs for a surplus record
POST	/api/surplus/{id}/respond	NGO accepts or rejects a surplus alert
GET	/api/reports	Consumption, surplus and waste-reduction reports

5.1.4 TEST PLAN AND VERIFICATION

Testing will be carried out at three levels, as listed in Table 5.4. Unit tests will cover the vote rules, the surplus and waste-rate calculations and the distance search. Integration tests will cover the calls between the PHP backend and the ML service. End-to-end tests will follow a vote through prediction, surplus entry, NGO alert and pickup confirmation.

Table 5.4. Planned tests

<i>Level</i>	<i>What is tested</i>	<i>Tool</i>
Unit	Duplicate vote rejection, deadline check, surplus and waste-rate maths, haversine distance	pytest, PHPUnit
Integration	PHP to FastAPI prediction call, database transactions, notification delivery	Postman, pytest
End to end	Vote to prediction to surplus to NGO alert to pickup confirmation	Manual and scripted

5.2 DISCUSSION

The design separates the web application from the machine learning service, keeps one database as the single source of truth, and closes the loop between prediction and the actual result. These choices make the system easy to build in stages and easy to test. The expected benefit is a lower waste rate, measured by equation 4.10 once real data is available, and a regular flow of surplus food to NGOs. Several risks must be watched. The prediction depends on the quality of the data, and a new cafeteria will have little history at first. Student votes may be unreliable, which is why the model learns from the actual attendance. Food safety rules limit what can be redistributed. Finally, any accuracy figures that will be obtained from synthetic data cannot be taken as proof of accuracy on real cafeteria data, and they will be reported with that clear statement.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This report presented the analysis and design of the Intelligent Food Waste Prediction and Redistribution System. The system starts with student votes, joins them with historical consumption, weather and calendar data, predicts the demand with XGBoost and Random Forest models, and helps the cafeteria decide the quantity to cook. After the meal it records the surplus, finds the nearest registered NGOs and alerts them so that edible food reaches people in need.

The strongest part of the design is the closed loop: votes lead to a prediction, the actual consumption is recorded, and the surplus is passed on, while every recorded value improves the next prediction. The main challenges are the quality and quantity of the data in the early weeks and the need to follow food-safety rules in redistribution. The work supports the goals of Zero Hunger, Responsible Consumption and Production, and Climate Action. The next phase will implement the modules, train and evaluate the models, and measure the reduction in food waste.

6.2 FUTURE WORK

The system forms a good base for a campus food-management platform. The following improvements are recommended.

- Extend the prediction from breakfast and lunch to dinner, snacks and special events, and add per-dish demand prediction.
- Use machine vision to measure plate waste automatically, as studied in [13], and use IoT sensors for temperature and food-condition monitoring [12].
- Add a rating and reminder feature for students, and a no-show penalty or reward scheme to improve the reliability of votes.
- Optimise NGO selection with capacity, vehicle availability and past response time, using ideas from food-rescue optimisation research [1], [16].
- Add a mobile application with push notifications for students, staff and NGO volunteers.
- Add carbon and cost reports that estimate the emissions and money saved by every kilogram of food that is not wasted.

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Intelligent Food Waste Prediction and Redistribution System Using Machine Learning

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Abstract—Food waste in college cafeterias can be attributed to student attendance uncertainty, inaccurate meal demand estimation, and the lack of organized surplus redistribution. Traditional methods of food preparation rely either on experience or rough estimates of student attendance, leading to either surplus or insufficient food preparation. This article describes an innovative system of predicting and redistributing food waste in college cafeterias. The system incorporates student voting, machine-learned demand estimation, and surplus monitoring and NGO coordination mechanisms to help reduce surplus food. Students can vote on a web portal for their breakfast and lunch needs. These results, past consumption data, customer numbers, food preparation, seasonal adjustments, holidays, and events can be used to predict demand. The estimated number is compared to the number of meals planned to be prepared to estimate the amount of surplus food. After meals, cafeteria workers can monitor the amount of surplus food recorded in the system. Finally, the application can notify registered nearby NGOs in case of edible surplus to collect and redistribute food to those in need. The described system had food demand estimation, voting, surplus tracking, NGO identification, and collection monitoring subsystems. While the paper demonstrated the predicted number of meals and voting results, there are no numerical results to see how well the model performed or reduced food waste in practice. More research will be required to evaluate the system's practical performance with real cafeteria demand data. The paper's contribution is that it provides the framework for tying food demand estimation to a surplus redistribution system and NGOs.

Index Terms—Food waste management, machine learning, demand forecasting, cafeteria management, food redistribution, surplus food, sustainability, NGO coordination.

I. INTRODUCTION

Food wastage in educational institutions emerges as a significant issue due to the lack of synchronization between the supplied quantity of food and the consumption demand

of the students. College and university messes suffer from the inaccuracy of meal planning as they prepare the food keeping in mind only approximated figures of attendance. It creates problems including wastage of food, utilization of extra resources, and additional expenses for the messes along with adverse environmental effects. Hence, there is a need for an efficient and effective system for the better estimation of meal demand and management of food resource planning. This paper proposes the FoodPulseAI project, a smart food-demand management and student meal-voting system, for educational institutions. The system allows students to provide their voting preferences about the meals through their web interface. Students would provide their voting input for their respective breakfast and lunch timings; thus, helping the messes in having an exact estimation of the expected number of consumers. The website will keep track of all the students' information and voting preferences in a database. It will also provide a dashboard for the messes for tracking the voting data of students and managing all food-related activities of the institution. The voting data of students can be helpful in determining the meal patterns for the mess. Through the collective participation of all students, the project will ensure decreased wastage of food. The system will keep track of managing meal schedules, surplus foods, and other relevant activities related to food. Thus, through smart technologies, the paper proposes a food management system for educational institutions. Keeping in mind the benefits of estimating the needs of the students, the project will help the messes in transforming their traditional planning system towards a modern and smart one. Consequently, the primary focus of the development of this project is to provide a digital platform between the students and messes for effective food management. It will help in ensuring a sustainable use of resources at college and

university levels and provide a better mechanism for managing food resource planning.

II. LITERATURE SURVEY

Food waste has become an essential issue in institutional and large-scale food-service settings, where miscalculations of meal demands can lead to excessive amounts of food being prepared and wasted, increasing resource wastage and operational expenses. Recent studies have investigated the use of machine learning, digital platforms, demand forecasting, food-rescue systems, and intelligent redistribution to deal with these difficulties. Shi et al. [1] introduced a bandit data-driven optimization strategy for crowdsourcing food rescue platforms, demonstrating how data-driven optimization can enhance food-rescue operations and volunteer engagement. Their work stresses the value of using continually generated platform data to increase the effectiveness of food recovery operations. In a similar vein, Wan et al. [2] proposed a machine-learning framework for compost maturity prediction. Using feature analysis and prediction modeling, their research demonstrated how machine learning can be utilized to automate food-waste processing and assist intelligent waste-management systems. Suresh et al. [3] introduced the ZeroHunger digital solution that employs a combination of AI/ML, web application, Firebase, food quality assessment, and donor-recipient coordination to coordinate food donations with organizations in real time. Kumar et al. [4] investigated food-waste prediction using Random Forest, geolocation, and automated notifications to connect food donors to nearby NGOs. These studies have demonstrated that intelligent prediction and digital coordination can aid both food-waste reduction and redistribution. Kim and Yoon [5] investigated the use of machine-learning techniques to predict food-waste quantities in military dining facilities. Their work used environmental factors and meal composition to indicate that machine-learning models can be applied in large-scale food-service settings to estimate waste quantities. Faezrad et al. [6] investigated university dining systems and introduced an Artificial Neural Network (ANN)-aided model using meal reservation data and student show/no-show behavior to predict demand and reduce surplus food. Eriksson et al. [7] investigated machine-learning-based guest attendance prediction in the context of Swedish public catering. Their work indicated that predicting expected attendance can assist in more appropriate food preparation and thus reduce food waste. Rodrigues et al. [8], investigated the short-term food-demand forecasting using historical catering data, indicating the potential that machine-learning-based demand forecasting has for the estimation of meal requirements and waste-reduction strategies. Jung and Kim [9] introduced machine-learning predictive models and a web application for university cafeteria attendance, using XGBoost and Random Forest models to predict breakfast, lunch, and dinner attendance. This work is particularly relevant to educational institutions as it demonstrates how cafeteria attendance prediction can be enhanced using a web-based system. Turker [10] introduced a data-driven approach to campus dining, emphasizing the use of

demand prediction and sustainability. The work highlighted the potential that campus-related data analytics have for enhancing food-planning decisions and reducing food waste.

In addition to demand prediction, computer-vision approaches have been investigated for the detection of food waste. Li et al. [11] introduced a machine-vision based approach for the automatic identification of food waste in university cafeterias using deep learning and plate-image recognition. This demonstrates the applicability of artificial intelligence not only in the field of food-demand prediction but in the measurement of actual food-waste as well. Agmapisarn et al. [12] investigated surplus-food redistribution through digital platforms, emphasizing real-time food listings and collaboration with charitable organizations. Their work demonstrates the key role of accessibility and real-time information in effective surplus-food redistribution. Shivani S. S. and Usha M. [13] introduced a web-based food donation and redistribution system using a centralized database, location tracking, and automated notifications. The system aims to enhance the coordination between food donors and NGOs, facilitating timely collection of surplus food. Devika et al. [14] expanded on this concept and introduced an IoT-enabled surplus food rescue platform using sensors, Firebase, web technologies, and NGO notifications. These systems demonstrate the importance of the integration of monitoring and communication technologies to food-rescue operations. Agmapisarn et al. [15] emphasize on the role of mobile and digital platforms in facilitating surplus-food sharing. Their work demonstrates the potential that real-time food listings and sustainability-oriented features have for encouraging food redistribution and food-waste reduction. Hübner et al. [16] investigated the utilization of machine-learning-based demand forecasting in combination with life-cycle environmental assessment in a bakery case study. Their findings demonstrate that enhanced demand forecasting can reduce unnecessary food returns while benefiting the environment. Food redistribution can be viewed as an optimization problem as well. The integrated foodbank network design discussed in [17] employs Mixed-Integer Programming to coordinate donors, foodbanks, recovery agencies, and beneficiaries. The case study presented here indicates the potential that network-level optimization has on the enhancement of the efficiency and equity of food redistribution services. Oroski and da Silva [18] through a review of food-waste-reducing platforms, have emphasized the rising role of digital food-sharing platforms in connecting surplus-food suppliers with recipients, monitoring food waste, and generating useful information for the purpose of food recovery. Murthy et al. [19], introduced a web-based surplus-food redistribution system that records surplus-food information and generate location-based alerts for NGOs, volunteers, and collection organizations. This demonstrates the usefulness of online coordination in the enhancement of the speed and efficiency of surplus-food collection. Turker [20], further emphasized on the utilization of data-driven campus dining, employing machine-learning regression and campus attendance/dining data in the prediction of daily meal demand.

III. PROPOSED SYSTEM

The proposed FoodPulseAI is an intelligent web-based food-demand management system, aimed at reducing the food wastage in campus cafeterias by predicting the daily meal requirement. The system exploits the real-time student meal voting, historical cafeteria data, contextual information, machine-learning based demand prediction, administrative monitoring and surplus-food redistribution in a combined framework. Unlike the ordinary cafeteria planning methods that base on the estimated attendance or historical consumption, FoodPulseAI captures the students' meal requirements before food preparation. Students can independently indicate their requirement of breakfast and/or lunch, while the administrators can monitor the collected responses and use the resulting demand information for food-planning decisions.

The proposed system comprises the student-side interface, administrative interface, backend processing layer, centralized database, demand prediction module, and surplus-food management module.

A. Student Module

The student module offers registration, authentication, and meal-voting functionality. A new student can register using the required student details and then access the system using valid credentials. After a successful login, the student is redirected to the meal-voting interface. The student is not expected to select individual food items. Instead, the system addresses the requirement of each meal.

The voting interface offers:

Breakfast - Yes/No
Lunch - Yes/No

Breakfast and lunch are considered as independent voting activities. Therefore, a student can submit a breakfast requirement without submitting a lunch response, or vice versa. This simplifies the participation of students while generating meal-specific demand information.

B. Real-Time Meal Demand Collection

Each student response is associated with the student's identity, meal type, voting date, and submission information. The response is validated by the backend before being stored in the centralized database.

The number of students requesting meal m is calculated as:

$$V_m = \sum_{i=1}^N I(v_{i,m} = \text{Yes})$$

where V_m represents the current demand signal for meal m , N represents the participating students, and $I(\cdot)$ is an indicator function.

C. Administrator Module

The administrator module offers centralized control over the food-management process. The administrator can access the dashboard to monitor student participation and meal requirements. The dashboard presents the current breakfast and lunch voting information, allowing administrators to monitor changes in expected demand.

D. Voting-Time Management

The administrator can define the permitted voting period. The backend verifies the configured date and time before accepting a student response. This prevents voting outside the permitted period and provides administrators with control over the data-collection process.

E. Machine-Learning Demand Prediction

The proposed system combines the real-time student voting with historical cafeteria data and contextual factors for demand prediction.

Historical data can include:

Prepared quantity
Consumed/sold quantity
Surplus quantity
Meal type
Date

$$X = [V_m, D, M, T, R, H, E] \quad (1)$$

where V_m is the current student vote count, D is the day of the week, M is the month, T is the temperature, R is the rainfall, H is the holiday indicator, E is the examination indicator.

$$\hat{Y} = f(X) \quad (2)$$

where f represents the trained prediction model and $\hat{Y}$ represents the estimated meal demand.

F. Food Preparation Recommendation

The predicted meal requirement is converted into a recommended preparation quantity. To compensate for prediction uncertainty and reduce the potential for insufficient food, the proposed system applies a 10 percent safety factor.

$$Q_{\text{prep}} = \hat{Y} \times 1.10 \quad (3)$$

For example, if the predicted demand is 100 meals, the recommended preparation quantity is:

$$Q_{\text{prep}} = 100 \times 1.10 = 110 \quad (4)$$

The safety factor can be adjusted based on institutional requirements and observed prediction performance.

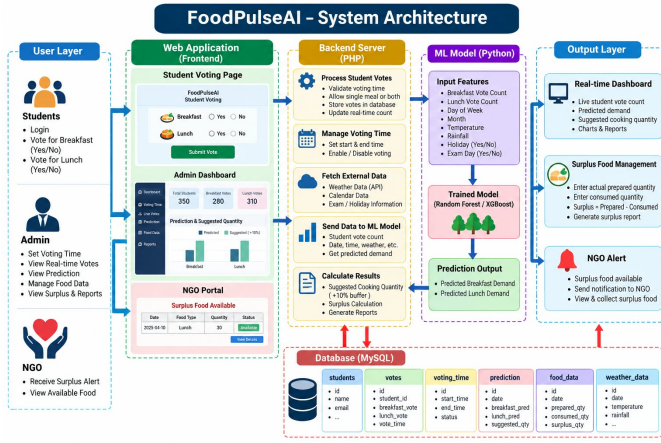


Fig. 1. System Architecture

G. Surplus Food Management

After food preparation and distribution, actual prepared and consumed quantities can be recorded. The surplus quantity is calculated as:

$$Q_{\text{surplus}} = \max(0, Q_{\text{prepared}} - Q_{\text{consumed}}) \quad (5)$$

If the resulting surplus is greater than zero, the system identifies the meal as surplus food. This creates the feedback mechanism between demand prediction and actual cafeteria outcomes. The recorded consumption and surplus information can then become part of the historical dataset for future demand prediction.

H. NGO and Redistribution Module

The NGO module keeps information about organizations that can support surplus-food redistribution. When suitable surplus food is identified, administrators can coordinate with registered NGOs.

Surplus Detection → NGO Identification → Notification
→ Collection → Redistribution (6)

This ensures that the proposed system does not focus only on preventing excess food preparation but also offers the mechanism for managing unavoidable surplus food.

IV. METHODOLOGY

The FoodPulseAI methodology is characterized by a series of steps of collecting, preprocessing, predicting, evaluating, and using meal-demand data. It will be based on student votes in real-time and historical cafeteria data.

A. Data Collection

The first step is to collect data. In this example, it will be done in real-time based on student votes. The authenticated students will be able to vote if they need breakfast and/or lunch. Their votes will be stored in a database along with the student ID, meal type, and date. On the other hand,

the historical data will contain the cafeteria data, such as food prepared, consumed, and surplus, meal type, and the date. Other contextual variables are the day, month, holiday, examination, temperature, and rainfall.

B. Data Preprocessing and Feature Engineering

The second step is to preprocess and engineer the data. The database will be cleaned and formatted in the machine-readable form. The variables will be converted into features, namely the day, month, holiday, examination, temperature, rainfall, the number of students who voted for a particular meal as the input feature, and the total food demand or consumption as the target variable.

C. Machine-Learning Model Development

The study will consider two (2) machine learning models, namely, Random Forest Regression and XGBoost Regression to predict demand for meals. The historical data will be used to train the model. It will use the input features and the target variable identified

D. Model Evaluation

The next step is to evaluate the model. The study will generate the Mean Absolute Error (MAE) given by the equation below:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (7)$$

where Y is the actual demand and $\hat{Y}$ is the predicted demand. The model with the lowest error will be used for the final set of data.

E. Demand-based Food Preparation

Using the predicted demand, the food preparation quantity will be calculated as: followed by equation (3)

This will give a safe percentage of 10 for food preparation with some leeway for errors.

F. Surplus Detection and Redistribution

After food distribution, the study will calculate the surplus using the following equation (4). The information will be stored in the system and shared with the NGO coordination module. The surplus detection process will involve steps 1 through 5 above. The following workflow will be followed:

Surplus Detection → NGO Selection → Notification → Collection → Redistribution

The actual consumption and surplus may also be stored as part of the historical data set for future demand analysis.

G. Real-Time Demand Updating

The process will be repeated daily during active voting to update the demand based on the latest student preferences. This can be represented by the following equation:

Student votes → Database → Feature Generation → ML Model → Demand Prediction → Preparation Recommendation

The above workflow will ensure that FoodPulseAI uses real-time student demand and historical demand patterns to inform food planning and preparation.

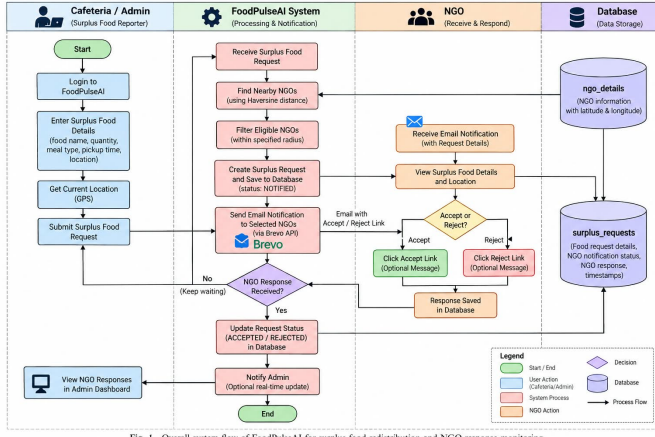


Fig. 2. Flow Chart

V. RESULTS AND DISCUSSION

The FoodPulseAI system allows for a complete collection of students' meal requirements, estimating the demand for food at the cafeteria, and managing surplus. The results of these processes are discussed in terms of student voting, demand prediction, food preparation, and surplus management.

A. Student Voting Results

The student voting function allows students to choose their preferences regarding the breakfast and lunch individually. For the voting process, students are required to log in to the system using the authentication module in order to make their choice. The results of the voting are collected and stored in a common database, separated into the data for breakfast and lunch.

Meal Type	Voting Options	Result
Breakfast	Yes / No	Breakfast demand count
Lunch	Yes / No	Lunch demand count

Table 1: Meal Demand Voting

The number of votes "Yes" can indicate the demand for each particular meal. Therefore, using this method, administrators can control the students' needs without estimating them manually.

B. Demand prediction results

The information about votes can be used along with the past data of the cafeteria demand and specific features to predict future demand. For this purpose, Random Forest and XGBoost Regression algorithms can be implemented.

Model	Input Features	Target	Evaluation
Random Forest	Votes, historical and contextual data	Meal demand	MAE
XGBoost	Votes, historical and contextual data	Meal demand	MAE

Table 2: Demand Prediction

The Mean Absolute Error (MAE) is used to compare the prediction models. A lower MAE indicates that the predicted demand is closer to the actual consumption. Therefore, the model with the lower MAE can be selected for demand estimation.

C. Food Preparation Results

The anticipated demand is modified into a recommended quantity of preparation using a 10percent safety factor. This will give the cafeteria staff enough but limited food for preparation with some leeway for the expected demand.

D. Surplus Food Management

The surplus is then generated by comparing the served quantity with the prepared quantity. This surplus is recorded and reported as the amount of food wasted while preparing meals and is reportable to the NGO coordination module.

Stage	Information
Food Prepared	Total quantity prepared
Food Consumed	Quantity consumed
Surplus Detected	Remaining quantity
NGO Selection	Suitable registered NGO
Notification	Surplus availability communicated
Redistribution	Surplus transferred for further use

Table 3: Food Management

E. Discussion

The analysis shows that the FoodPulseAI connects the current student turnout, which feeds back into the cafeteria demand forecasting and management module, and the surplus management. On the one hand, the student voting provides up-to-date information on the demand, while the historical and contextual data provide the basis for the prediction. On the other hand, the surplus module also offers an additional benefit of a surplus management system.

VI. CONCLUSION AND FUTURE SCOPE

A Food-Waste Prediction and Management Model For School Canteens A Solution To The Needs Of Student Using Analytics Essay A solution to the needs of students should be developed by considering ways of addressing the requirements and reducing wastage in institutes. Using the proposed solution, students will virtually provide their demands for breakfast and lunch. The two responses will help the authorities to make decisions on the number of meals to prepare in a day. Besides, it will be possible to predict demand by considering the submitted request and history of the cafeteria. For example, using random forest regression and XGBoost Regression algorithms, one can predict demand based on the gathered data and current requirements. The results can then be compared with a threshold to get an approximate target quantity to prepare. The proposed approach will be innovative by considering the student data to come up with a new way of estimating the meal demand in a cafeteria. As a result, the administration will address the problem of excess preparation over the expected quantity of students. Therefore, it will help reduce wastage and use of electricity in a school. Another strength of the proposed solution is that it will help manage the surplus. At the time of preparation, the system will support the management in recording the excess meals as the number of servings to students. Thus, the suggested solution will solve the prediction and management of surplus in a cafeteria. A food-waste management system in a cafeteria benefits tremendously from a prediction model. There is a need for administrative staff to develop an appreciation of the power of historical data in a cafeteria. First, it is important to record all meals prepared and served to help create a statistical model for the new system. Secondly, the administrator should consider recording the data on holidays, examinations, and class attendance schedules. Besides, weather conditions are also a good predictor of the demand for meals in a cafeteria.

In the light of future developments, FoodPulseAI can be developed further by using a wider range of historical demand data sets to increase accuracy of the predicted demand. More advanced machine-learning and deep learning techniques can also be explored and compared with the proposed algorithm other than the proposed Random Forest and XGBoost. Dynamic factors such as weather, attendance rates, examinations, holidays, and campus events can be included in the prediction system.

The system can also be expanded by integrating a mobile application to support student voting and administrative supervision. Automated student reminders can also be included which can inform students of voting periods and alert cafeteria staff of significant demand changes. Similarly, the system can be programmed to automatically send notifications to registered NGOs in the event of an occurrence of surplus food that can be redistributed.

Lastly, FoodPulseAI can implement a self-actualizing system that updates the history demand database with consumption and surplus figures after every meal cycle, essentially

teaching the system from its previous errors. In addition, the system can be expanded at the department, hostel, cafeteria, and institutional level to monitor and compare food wastage and demand figures generating valuable insights to enhance food management practices.

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Track Name: ICCIDS2027

Paper ID: 124

Paper Title: FoodPulseAI-Intelligent Food Waste Prediction and Redistribution System Using Machine Learning

Abstract:

Food waste in college cafeterias can be attributed to student attendance uncertainty, inaccurate meal demand estimation, and the lack of organized surplus redistribution. Traditional methods of food preparation rely either on experience or rough estimates of student attendance, leading to either surplus or insufficient food preparation. This article describes an innovative system of predicting and redistributing food waste in college cafeterias. The system incorporates student voting, machinelearned demand estimation, and surplus monitoring and NGO coordination mechanisms to help reduce surplus food. Students can vote on a web portal for their breakfast and lunch needs. These results, past consumption data, customer numbers, food preparation, seasonal adjustments, holidays, and events can be used to predict demand. The estimated number is compared to the number of meals planned to be prepared to estimate the amount of surplus food. After meals, cafeteria workers can monitor the amount of surplus food recorded in the system. Finally, the application can notify registered nearby NGOs in case of edible surplus to collect and redistribute food to those in need. The described system had food demand estimation, voting, surplus tracking, NGO identification, and collection monitoring subsystems. While the paper demonstrated the predicted number of meals and voting results, there are no numerical results to see how well the model performed or reduced food waste in practice. More research will be required to evaluate the system's practical performance with real cafeteria demand data. The paper's contribution is that it provides the framework for tying food demand estimation to a surplus redistribution system and NGOs.

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